Supermarket Sales Analysis

Data Analytics Project

1. Problem Statement

This project aims to study supermarket sales data and uncover useful insights about products, branches, categories, customers, payment methods, and ratings.

2. Dataset

The dataset holds 500 sales transactions. It covers product, branch, city, customer type, quantity, price, payment method, rating, and sales.

3. Steps Followed in the Analysis

1. Collect and load the CSV dataset.
2. Check the data for missing or incorrect values.
3. Calculate Sales = Quantity × Unit Price.
4. Group and summarize the data using totals, counts, and averages.
5. Build charts to compare the results.
6. Apply the results to make business decisions.

4. Analysis and Findings

Which product earns the highest sales?

Cheese earned the highest sales at ₹27,906.30.

Which branch performs the best?

Branch C (Mumbai) performed the best, with sales of ₹72,469.45.

Which category sells the most?

Beverages recorded the highest sales at ₹56,108.24.

Which payment method is used the most?

UPI was the most used payment method, with 127 transactions.

Do Members spend more than Normal customers?

No. Members averaged ₹483.14 per transaction, while Normal customers averaged ₹497.07.

What is the average customer rating?

The average customer rating was 3.99 out of 5.

5. Business Decisions

The supermarket can maintain more stock of fast-selling products and categories, look into what drives Branch C's strong performance, keep supporting UPI payments, and improve customer service to raise ratings. Customer spending patterns can also help in planning membership offers.

6. Conclusion

This analysis shows how sales data can be turned into simple insights that help a supermarket understand its performance and make better decisions.